

Abstract 3699 – Table 1: OAR dosimetric results of subclinical dose de-escalated radiotherapy for HPV+ OPC

T1-2/N1-2b And T3-4/N0 Patients						
	N	Mean (Gy)	Median (Gy)	SD	Min (Gy)	Max (Gy)
Ipsilateral Parotid Mean Dose	101	20.3	19.3	7.3	2.7	61.4
Contralateral Parotid Mean Dose	101	8.6	8.3	2.6	1.2	20.2
Contralateral Submandibular Gland Mean Dose	97	28.9	25.7	9.7	6.3	65.0
Oral Cavity Mean Dose	99	38.8	38.4	10.6	6.3	93.4
Larynx Mean Dose	101	22.2	21.5	10.7	1.1	69.9
N2c Patients						
Right Parotid Mean Dose	69	17.6	16.6	8.7	5.9	65.9
Left Parotid Mean Dose	69	17.6	16.6	67.0	4.7	39.1
Right Submandibular Gland Mean Dose	65	56.6	57.6	14.8	24.5	102.4
Left Submandibular Gland Mean Dose	65	54.9	57.8	16.4	22.0	103.3
Oral Cavity Mean Dose	69	44.4	42.1	11.8	16.1	87.7
Larynx Mean Dose	68	24.8	25.6	10.1	5.9	54.7

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3700

Gamma Knife Stereotactic Radiosurgery with a Reduced Peripheral Dose: Outcomes in Uveal Melanoma Patients

S. Mishra,¹ A. Gao,² L. Kollitz,² G. Sodhi,³ P. Tang,⁴ Y. Watanabe,¹ A. Venteicher,⁵ D. Koozekanani,⁶ C. Ferreira,¹ E. Domingo-Musibay,⁷ J. Yuan,⁸ M.A. Reynolds,¹ and K.E. Dusenbery¹; ¹University of Minnesota: Department of Radiation Oncology, Minneapolis, MN, ²University of Minnesota, Minneapolis, MN, ³Retina Consultants of Minnesota, Minneapolis, MN, ⁴Storm Eye Institute, Charleston, SC, ⁵University of Minnesota: Department of Neurosurgery, Minneapolis, MN, ⁶University of Minnesota: Department of Ophthalmology and Visual Neurosciences, Minneapolis, MN, ⁷Allina Health, Minneapolis, MN, ⁸Department of Radiation Oncology, University of Minnesota Medical School, Minneapolis, MN

Purpose/Objective(s): Uveal melanoma (UM), the leading primary intra-ocular malignancy in adults, often falls outside the treatable scope of plaque brachytherapy for large or juxta-papillary tumors. This study evaluates a dose-reduced non-invasive Stereotactic Radiosurgery (SRS) protocol amidst varied SRS practices and doses (18-45 Gy) and aims to standardize treatment and assess its effectiveness and safety for comparable local control to plaque brachytherapy.

Materials/Methods: In this IRB-approved, retrospective non-comparative case series, we evaluated clinical and radiosurgical data from 17 UM patients treated with GK SRS and surgical fixation of extraocular muscles from 2020 to 2023. Treatment parameters included tumor volume, peripheral dose, tumor coverage, gradient index, and maximum doses to the lens and optic nerve. Treatment efficacy was assessed through changes in primary UM apical height pre- and post-SRS to quantify local control. Safety outcomes include visual acuity, cataract progression, and the incidence of radiation retinopathy, glaucoma, neovascularization, and enucleation.

Results: 17 UM patients (65% male, 59% left-sided) underwent GK SRS. Mean tumor volume was 0.964 cm³ (± 0.446 cm³), with 35.3% large and 64.7% medium in size, per COMS classification. A median peripheral dose

of 24 Gy (range = 24-25 Gy) was administered, achieving 98.9% (± 1%) tumor coverage with gradient index 2.89 (± 0.19). At mean follow-up of 27.8 months (± 6.9 months), we noted a 24.9% decrease in tumor apical height at 1 year (p = 0.00015) and a 30.5% decrease at the last follow-up (p = 0.00013), compared to baseline, with an overall 94.4% local control. Visual acuity was stable or improved in 64.7% of patients at 1 year, with 52.9% having >20/200 vision at last follow-up. 53% experienced radiation retinopathy and 42.9% of patients without prior cataract surgery had cataract progression. Glaucoma developed in 23.5% of patients, with no cases in the first year. 35.3% had neovascularization, of whom half required enucleation.

Conclusion: Our study confirms GK SRS as an effective alternative to enucleation for UM patients unsuitable for brachytherapy or for those desiring to avoid surgery. Despite treating medium and large UM with a relatively low peripheral dose of 24 Gy, local control was high and comparable to COMS report 19, with most patients avoiding legal blindness. However, patients still had high risk of cataract progression, radiation retinopathy, and glaucoma, with neovascularization often preceding enucleation. Based on our results, we propose a 24 Gy GK SRS protocol as a viable treatment option for UM, striking a balance between effective tumor control and visual function preservation.

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3701

Risk Factors for Developing Radiation Related Late Toxicities after Treatment for Head and Neck Cancers and Impact on Survival

J. Mohan Mathew,¹ J. Ringash,¹ J. Su,² M. Woodside,¹ S.V. Bratman,¹ J. Cho,¹ E. Hahn,¹ A. Hosni,¹ A.J. Hope,¹ J. Kim,¹ A. McPartlin,¹ B. O'Sullivan,¹ C.J. Tsai,¹ J. Waldron,¹ J. Kwan,¹ A. Spreafico,³ D. Goldstein,⁴ S.H. Huang,¹ and P. Wong¹; ¹Department of Radiation Oncology, Princess Margaret Cancer Centre, University of Toronto, Toronto, ON, Canada, ²Department of Biostatistics, Princess Margaret Cancer Centre, University of Toronto, Toronto, ON, Canada, ³Department of Medical Oncology, Princess Margaret Cancer Centre, University of Toronto, Toronto, ON, Canada, ⁴Department of Otolaryngology-Head & Neck Surgery, Princess Margaret Cancer Centre, University of Toronto, Toronto, ON, Canada

Purpose/Objective(s): Identification of severe radiation-related late toxicity (RLT) risk factors in patients treated for head & neck cancers (HNCs) could guide mitigation strategies & follow-up routines. Impact of RLT on long term overall survival (OS) is also not well studied in HNC survivors. We investigate potential predisposing factors & association with long term outcomes for patient with severe RLTs.

Materials/Methods: We performed a retrospective case-control study, using data from our institutional prospective head & neck anthology of outcome system & case registry of the Adult Radiation Late Effect Clinic. We included HNC patients treated from 2003 – 2020 who received radiation (RT) ≥ 50 Gy without any treatment failure. Those with documented RTOG grade 3-4 RLT (≥ 6 months post-RT) were classified as cases. Those without documented grade 3/4 RLT & follow-up of ≥ 2 years served as controls. Cases could have one or more RLTs & date of RLT was recorded at the 1st occurrence. Baseline characteristics were compared between cases & controls. Actuarial rates of LT & OS were calculated using competing risk & Kaplan-Meier methods, respectively. Multivariable logistic regression analyses (MVA) were performed to identify risk factors (odds ratio, OR) for RLT & prognostic factors (hazard ratio, HR) for OS.

Results: Our analysis included 4797 patients (1048 cases & 3749 controls), with a median follow-up of 5.5 years. The most common RLTs observed were: osteoradionecrosis (n = 355), dysphagia or feeding tube dependence